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Challenges and possible severe legal consequences of application users identification from CNG-Logs

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ABSTRACT

Due to the increase in the use of Internet-connected devices and the limited number of available Internet Protocol Version 4 (IPv4) addresses, a system necessity that would share the same Internet Protocol (IP) address among multiple end-users emerged. These systems that will allow end-users to share the same IP address are commonly called Carrier-Grade Network (CGN) Address Translation (CGNAT). Even though the laws may vary among different countries, many of the current nations impose Internet Service Providers (ISPs) to keep internet users' activity as logs. Usually, law enforcement is provided with the exact date, public IP address, and the public port number to detect cybercriminals. Unfortunately, though, when reverse-tracking CGN logs methodology is followed, while the precision of actor detection decreases, the false positive rate increases. Therefore, this methodology has significant issues and can cause innocent people to be treated as criminals. As a case study, this study investigated criminal cases in Turkey, where the defendants are accused of being part of a terrorist organization due to the use of a mobile chat application called ByLock. We explored incorrect usage of CGNAT records to detect users of a specific application and how it affected masses from a criminal perspective. In this paper, the dangers of such an approach will be investigated in-depth by providing an actual criminal case example from the ByLock cases. Furthermore, the paper also compares the reverse-tracking of CGNAT logs approach with the approach followed in another mobile chat application (EncroChat) in terms of cybercriminal mass investigations.

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1. Introduction

IPv4 is the fourth generation of IP, and it was defined in September 1981 (Postel, 1981). IPv4 uses 32-bit addresses; thus, its capacity is limited to 4,294,967,296 (2^{32}) addresses at maximum, but not all of the address space is useable (e.g., 10.x.x.x and 192.168.x.x IP blocks are reserved for internal use (McCarthy, 2017)). Due to such limited address capacity and the rapid increase of electronic devices that connect to the Internet, IPv4 exhaustion is inevitable. Furthermore, many of the major regional Internet address registrars, such as Asia–Pacific Network Information Centre (APNIC), Réseaux IP Européens Network Coordination Centre (RIPE), and American Registry for Internet Numbers (ARIN), already exhausted their IPv4 pool (McCarthy, 2017). The solid solution for IPv4 exhaustion is the transition from IPv4 to IPv6 (Levin and Schmidt, 2014). However, the transition to IPv6 requires many challenges to be solved since IPv6 is not backward compatible with IPv4 (Levin and Schmidt, 2014). Therefore, only less than 35% of the world still uses IPv6 to access Google (Google and "v6 Adoption St, 2020). For instance, Turkey has less than 0.02% IPv6 capability along with other 101 countries; meanwhile, India has the highest 70% capability among 240 countries (C and "v6 Capable Rate, 2020). Only India, Belgium, Malaysia, United States (US), and Mayotte have more than 50% IPv6 capability (C and "v6 Capable Rate, 2020). The temporary solution developed for IPv4 exhaustion is IP address sharing among multiple subscribers (Ford et al., 2011). The core of CGN technology is sharing a single IP address among multiple users (Huston, 2013) by forming NAT444 (extension technology of IPv4 (Donley et al., 2013)) environment on top of the subscriber's home/mobile Network Address Translation (NAT)

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system (Kuarsingh and Cianfarani, 2014). CGN systems are also known as Large Scale NAT (LSN) or NAT444 (Aitken, 2013). NAT technology is developed for the same private network-connected devices to connect the Internet by using the same global IP address. NAT translates private network IP addresses to public network IP addresses, as shown in Fig. 1 below, and it is usually called NAT44 (Keith, 2018). The main difference of CGN from NAT is the translation of private-to-private IP addresses, as shown in Fig. 2 below. Even though the details of the system can change according to the internal setup of ISPs, there are two types of IP address sharing among customers. The first method is port-multiplexing (Huston, 2013), which is a port-range solution (Ford et al., 2011). At this methodology, certain public port number and public IP address couple is assigned to specific customers at certain times. However, solely port-multiplexing is still inefficient for sharing a single IP address to multiple users. As a second method for improved IP address sharing, ISPs tend to use a combination of 5 variables: (a) transport protocol; (b) subscriber identifier (e.g., internal source address or tunnel endpoint identifier); (c) external source address; (d) external source port number; and (e) timestamp (Perreault et al., 2013), which is a CGN-based solution (Ford et al., 2011). More detailed information about IPv4, NAT44, and CGN can be found in (Aitken, 2013).

CGN systems have many issues for ISPs and subscribers (C and "About Carrier Gra, 2020), yet they are still used by the majority of the ISPs (more than 90% of the cellular networks use CGN systems (Richter et al., 2016)). Moreover, according to a very large-scale study conducted on the Internet, the number of CGN networks has increased significantly between 2014 and 2016 (Livadariu et al., 2018).

Among the unfavorable consequences, probably the worst for the subscribers would be the criminal charges based on faulty/erroneous CGN logs. ISPs are usually enforced by regularity laws to keep the logs of internet access requests (O'Reilly, 2017). However, due to the nature of CGN systems, the identification of individual subscribers from internet-access logs is challenging and error-prone. This challenge may become more severe and widespread as large-scale IP address sharing increases (Ford et al., 2011). For instance, according to an internal survey conducted by the European Cybercrime Centre among all European Union (EU) Member States cyber divisions in 2016, 90% of the respondents during their investigations have encountered errors regularly related to CGN technologies (Europol, 2016). Moreover, due to CGN related errors, they had to discontinue investigations, or they had to use more resources and time-consuming approaches to identify end-users accurately. This study explores the risks of using CGN logs in criminal cases. Moreover, it thoroughly explains how it may go dangerously wrong when CGN logs are used to do reverse-tracking to identify cybercrime suspects in criminal cases. Furthermore, it compares ByLock prosecutions with another similar-purposed service, EncroChat prosecutions executed by European authorities. Note that the focus of this research is mainly on the technical perspective of such criminal cases rather than a legal perspective. We sincerely hope that the results of this study will help forensics experts and judicial decision-makers on how to handle CGN logs to detect suspects.

2. Related work

The following studies have covered possible legal challenges at some scale and issues of large-scale IP address sharing and CGN systems. Olaf et al. proposed a better approach than carrier-grade NAT and asserted that CGNs are posing a security threat (Maennel et al., 2008). Huston has briefly analyzed the history of how individuals were identified by IP address logs and shared his opinions about the incoming or existing law enforcement legislations about CGN logs keeping (Huston, 2013). O'Reilly has discussed the main challenges of identifying individuals from CGN logs and proposed that server-side source-port logging awareness and the availability of the necessary tools should be increased (O'Reilly, 2017). Zander and Murray have investigated how widespread the utilization of large-scale address sharing on the Internet is (Zander and Murray, 2017). They found that even several thousands of people use only a single unique IP address in some of the ISPs. Dawadi et al. thoroughly summarized why the transition to IPv4 from IPv6 is mandatory, discussed the transition challenges, reviewed the different transition technologies, and discussed migration strategies from IPv4 to IPv6 (Dawadi et al., 2015). Ford and colleagues have discussed many of the problematic or challenging issues of large-scale IP address sharing among subscribers, such as port allocation, geolocation, fragmentation, traceability, security (Ford et al., 2011).

This study covers the severe criminal consequences that can be caused by IP address sharing and the erroneous implementations of CGN systems by ISPs much more thoroughly. Furthermore, it strengthens its arguments with case-studies from ongoing judicial trials in Turkey. Bocchi et al. investigated the effect of CGN usage on end-users when web browsing the Internet (Bocchi et al., 2015). They have found no significant effect of CGN from a user experience perspective. Nevertheless, as will be explained in this article, the main problem of CGN appears when its logs are used as evidence in criminal cases by authorities. Gokce has published an in-depth analysis of the ByLock investigations in Turkey and has briefly discussed how CGN logs were used to compose the list of ByLock

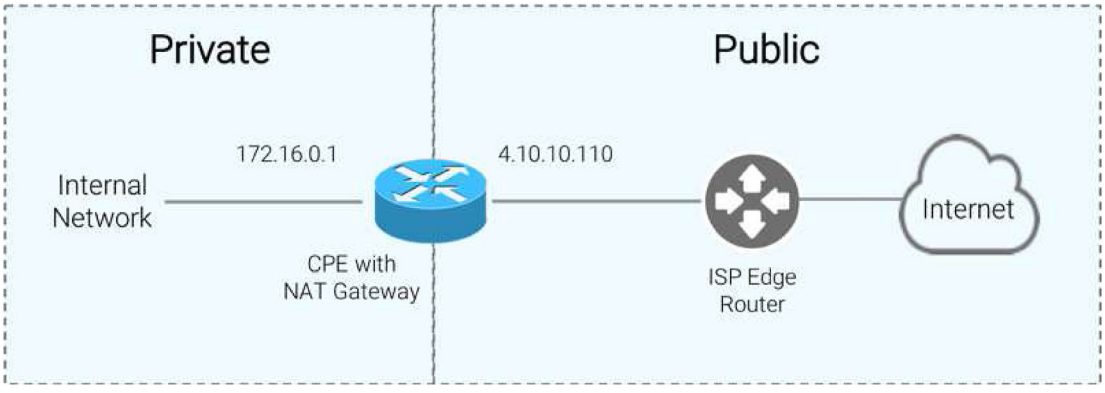


Fig. 1. Standard NAT – Translating private IP addresses to public IP addresses (Keith, 2018).

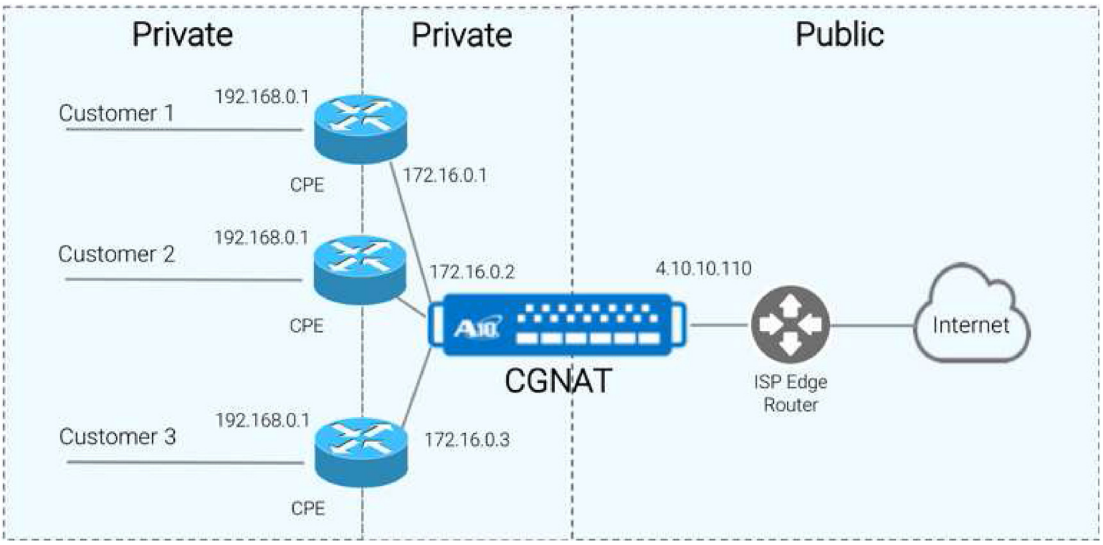


Fig. 2. CGN implementation of NAT444, private to private to public NAT (Keith, 2018).

users (Gokce, 2018). However, Gokce did not comprehensively discuss the technical details of the cases or the CGN problems, as in our research. Ozalp has published a case study about the ByLock prosecutions and asserted how CGN logs (they referred to as *internet traffic metadata*) were abused without delving into technical details of CGN systems (Ozalp, 2019). Škoberne et al. analyzed the tradeoff between ISP-level IP address sharing systems and classified them into nine different categories (Škoberne et al., 2014). However, they discussed the legal aspect of IP address sharing consequences very briefly. Ma et al. discussed why the logs of CGN systems have to be recorded and how the logging system can be improved with port mapping (Ma et al., 2012). However, their proposed system does not keep records of visited individual remote addresses; thus, it may not be helpful to judicial authorities in some legal cases. Perreault et al. published a standard requirements article for CGNs and briefly described how CGN logs should be stored (Perreault et al., 2013). Mackey et al. published an extensive white paper about incorrect search warrants, raids, arrests, and judicial cases caused by IP address resolutions (tracking back subscribers) in the US (Mackey et al., 2016). They extensively discussed how false IP address resolutions happen, how to avoid them possibly, how police forces and courts should behave according to decisions of the Supreme Court and the Fourth Amendment. However, they did not discuss CGN systems in detail. Moreover, no information related to the reverse-tracking methodology was provided in their research.

3. Communication over the internet and CGN logs

3.1. Parameters required to communicate over the internet

To understand how CGN logs are stored, first, we need to understand what is necessary for applications to communicate over the Internet. The internet-based software and services commonly use Transmission Control Protocol (TCP) or User-Datagram Protocol (UDP) sessions to communicate (Aitken, 2013). These sessions are uniquely individualized with the combination of five parameters (these parameters may vary if other than TCP or UDP protocols are used and a combination is called as socket) (Aitken, 2013): (a) source IP address; (b) destination IP address; (c) transport layer protocol (e.g., TCP or UDP); (d) source port number; (e) destination port number. If a CGN is used to trackback individual sessions to individual end-users from logs, it has to store all the above five parameters. In addition to these parameters, it has to store internal mapping parameters as well. These additional parameters usually include private IPv4 addresses and private ports of particular sessions and may change according to the service provider configuration. In (Škoberne et al., 2014), nine different types of CGNs' internal configuration are analyzed. Our provided example CGN system in Section 4.3 uses private IPv4 address, private port number, and time parameters to identify end-users uniquely. A comparison of the logging requirements of different network models is shown in Table 1 below.

For a CGN deployment, the logging requirements significantly increase compared to a static IPv4 deployment. According to a lab test done by (Donley et al., 2014), a typical CGN log requires 150 bytes on the hard drive. Moreover, they have found that an average US customer generates 33,000 sessions per day, which would require one million customers having ISP to store 1.8 petabytes per year just for logging. Furthermore, the study in (Donley et al., 2014) was done in 2013, and it is safe to assume that the number of average sessions would be more today due to more sophisticated and bandwidth-hungry applications. Besides, it is worth noting that such detailed logging brings no commercial benefits to the ISPs.

3.2. Number of customer limitations per public IPv4 address

Applications usually use multiple sessions to communicate over the Internet for various reasons, such as when browsing the websites downloading content in parallel to load pages faster. Each session uses a port on both the end user's device and the ISP's network. Moreover, the number of ports both TCP and UDP protocols can utilize is limited to 2^{16}, which is 65,536 ports, and some of these ports are reserved for specific uses (e.g., port 80 for HTTP). Thus, a client can use only some of the available ports. If the IP addresses assigned to the clients were not shared, the number of ports would be abundant in most cases for the requirements of the end-users. However, when a public IP address is shared among a certain number of customers, the ports are also shared. Therefore, a customer can only utilize a certain number of ports according to the service-provider configuration. When the number of ports (defines the number of sessions that the client simultaneously can have) shared among multiple customers is assumed to be equal, the number of sessions a client can have maximum is displayed in

Table 1
Comparison of logging requirements (Aitken, 2013).

Access Network	Service Provider Logging Requirements	Logging Requirements at Destination
Routed	None (fixed record of allocation)	Source IP address
Subscriber CPE with NAT44	None (fixed record of allocation)	Source IP address
Subscriber CPE with NAT44 (Dynamic public IP address)	Subscriber dynamic IP address (typically changes daily)	Source IP address + date and timestamp
CGN	Per session: • Date and time • Internal IP address (may be dynamic) • Internal source port number • External CGN source IP address • External CGN source port number	Per session: • Date and time • Source IP address • Source port number

Table 2
Maximum number of TCP sessions illustrative estimates (Aitken, 2013).

Scenario	Maximum Number of Sessions (TCP)	Comment
Routed	65,536 per node	Not all ports are available for client use
Subscriber CPE with NAT44	65,536 per subscriber	Not all ports are available for client use
CGN	Maximum set by carrier. Depends on CGN implementation and configuration.	Not all ports are available for client use. For example, well known ports.
Maximum 10 subscribers per IP address	6536 per subscriber	
Maximum 100 subscribers per IP address	653 per subscriber	
Maximum 1000 subscribers per IP address	65 per subscriber	

Table 2 below. Since there are some reserved ports, it will depend less on the configuration of the ISP in actual usage.

3.3. Deterministic NAT to reduce logging costs

Since some ISPs are bound to keep necessary logs for law enforcement while the cost of CGN logging is extraordinary due to the nature of how CGN systems work, deterministic CGN systems are proposed to reduce logging costs (Donley et al., 2014). In a regular CGN, any external IPv4 address and external port can be assigned to any customer at any given session dynamically from the available pool. However, this approach requires all five variables explained in Section 3.1 to be stored as logs to trackback individual subscribers from the public IPv4 addresses and the external port numbers. Therefore, in a deterministic CGN, instead of assigning internal/external IP addresses and port numbers dynamically from available pools, a custom range of port numbers and external IP addresses are assigned to the customers with a special mapping algorithm that determines the internal IP addresses and port numbers (Donley et al., 2014). The algorithm maps bidirectional both the external and internal IP addresses and port numbers, as shown below (Donley et al., 2014):

- External IP Address, Port Range = function_1 (Internal IP Address)
- Internal IP Address = function_2 (External IP Address, Port Number)

A more detailed example of how deterministic CGN translations are made can be found in (Platform and Release Supp, 2019).

There are three main disadvantages of deterministic CGN logging: (a) At any given time, the correct system configuration and algorithm parameters have to be stored to reverse-map CGN logs when necessary. Therefore, a new configuration log has to be generated and stored whenever any configuration parameter is changed. If configuration logs become inaccessible or lost, the reverse-mapping of individual subscribers from deterministic CGN logs would be impossible. Since configuration logs are generated and stored periodically, this may cause many sessions to become untraceable; (b) Unless dynamic port range is enabled, when a subscriber consumes all of the reserved port range, further connection attempts of the subscriber would fail. If additional ports dynamically get assigned to the client in such a case, the system must not assign any other clients to the reserved ports. If multiple clients get assigned to reserved ports at the same time, logs may be traced back to incorrect subscribers; (c) Since deterministic CGN configuration is more complex than port-assigning CGNs, it requires additional work to be done by system engineers to set up and optimize the system (Donley et al., 2014).

4. Possible dangers of using CGN logs in criminal cases

There are four main challenges of using CGN logs in criminal cases as evidence: (a) ISPs must store CGN logs detailed enough to identify individual end-users with high accuracy; (b) Integrity and precision of the CGN logs must be dependable enough to be used as evidence in criminal cases; (c) The relevant application servers have to keep the incoming port number of sessions in their logs; (d) The particular log of a session might have been generated by either the client's device itself or a third-party software without the customer's intent.

4.1. Necessary parameters to identify individual subscribers from CGN logs

When an activity performed by a computer system and a network involves a criminal act, it is classified as a cybercrime (Moore, 2011). The IP address and the timestamp are the key factors determining the prime suspects in the early stages of an investigation when a cybercrime is committed (Kao and Wang, 2009). However, when multiple users share the same public IP address to connect to the internet, the IP address with the timestamp no longer uniquely identifies the customer (Ford et al., 2011). Therefore, two solutions are proposed to overcome the problem (Ford et al., 2011): (a) Content providing servers (where a crime has been committed) keep the IP address, the source port number, and

the timestamp of all incoming connections; (b) ISPs do log the destination IP address of all requests. For the former one, the legal request to identify possible crime suspects has to include the IP address, the source port number, and the timestamp of the request. Moreover, the timestamps must be accurate since, in very dense CGNs, ports assigned to customers can change frequently, and the source port numbers of requests are crucial to determine end-users. The latter solution is imperfect and inferior to the former, because if multiple subscribers access the same server almost at the same time with multiple connections having protocols such as HTTP, it may not be possible to differentiate which subscriber is the actual source of the particular connection solely by depending on the timestamp and the destination IP address. Therefore, ISPs have to keep a record of destination IP address and port number as well in their CGN logs. However, destination IP address logging by ISPs is considered to cause privacy issues (Perreault et al., 2013). Furthermore, logging destination IP addresses and port numbers is an additional logging burden to ISPs. If neither approach is followed, it would be impossible for an ISP to identify a specific connection initiator solely based on the IP address and the timestamp.

4.2. Error-free CGN logging

A CGN system has to log all of the necessary parameters explained in Section 3.1 to keep error-free logs. If the CGN is deterministic, the error-free logging becomes even more complicated, with more variables that need to be set up correctly. Besides, logging brings no monetary benefit to ISPs; thus, ISPs keep logs only because local laws enforce them to keep. Furthermore, incorrect logs do not reduce the service experience of the consumers unless they accidently get into legal trouble due to erroneous logs. Because of all these reasons, ISPs may try to minimize logging costs without following the strict rules, which may lead to inaccurate logs. For example (Aitken, 2013), identified that the logging system of one of the mobile access providers in the United Kingdom was problematic. In (Aitken, 2013), one of the ISPs implied that it is infeasible to trackback to the individual customers from CGN logs; thus, they had no plan to store logs. For instance, the deterministic CGN system of F5 Networks had a bug (2013−2018) that prevented some of the CGN logs reverse-mapped to the individual customers correctly (Networks, 2018a). F5 Networks is a deterministic CGN-system provider for at least one of the major ISPs in Turkey (Headquarters, 2013). Considering how the CGN logs are stored, it would be impossible to fix previous logs after the discovery of such bugs in most cases. Moreover, the internally discovered and solved bugs are not likely to be disclosed to the public and get excluded from the logs of ISPs. This research also indicates that non-mobile (e.g., cable, DSL, fiber) internet providers in Turkey did not keep any session logs, at least between April 2014 and April 2016, which will be explained in Section 5.

The importance of erroneous logs from a legal point is the misidentification of the subscribers as we see in criminal investigations conducted by Turkish law enforcement after 2016. When an ISP is asked to provide CGN logs, the correctness of the logs is trusted to the ISP, and it is difficult to verify its integrity unless it has explicit errors externally. Incorrect logs have also been the case in the ByLock judicial trials in Turkey. ByLock was a mobile chat application, and whoever used the app was deemed to be a member of the Fethullahist Terrorist Organization (FETO) by law enforcement (Fox-IT, 2017). Thus, law enforcement started a mass investigation to identify the users of the ByLock application (Agency, 2016a) (Zorlu, 2018). For more information related to judicial trials about the ByLock cases, you can see (Gokce, 2018) and (Clegg QC and Baker, 2017), and more information about the technical details of the ByLock application can be found in (Fox-IT, 2017) and (Dulkadir et al., 2018). There have been several lists of ByLock users generated by law enforcement and intelligence agencies (News, 2017); however, this study, considers the latest list of ByLock users, which was prepared by using the CGN logs. Turkey's National Intelligence Organization (MİT) prepared a technical report about the ByLock service (Fox-IT, 2017) and sent it to the Ankara Chief Public Prosecutor's Office (ACPPO). The cover letter of the ByLock Technical Report of the MİT was later revealed (Aktaş, 2018). More details about this cover letter can be found on (Dulkadir et al., 2018), page 28. The cover letter explains how the list of ByLock users was prepared. Moreover, the trust of law enforcers in the data provided by the ISPs is described in the letter, as shown in Fig. 3 below.

The translation[1] of the red-underlined sentence shown in Fig. 3 is as follows: “The operators who provide services in the electronic communication sector have to provide the information and documents that they are required to submit by the Legislation on the Information Technologies and Communication Authority, which they are primarily subjected to, in a “correct” and “complete” way.”

4.3. An actual example of flawed CGN logs used as evidence in a judicial trial

Even though the correctness of the CGN logs can only be ensured and verified by the internal team of ISPs, certain cases proved that the data sent by ISPs are incorrect and flawed. For example, Fig. 4 below shows an actual example of CGN query logs requested by the Ankara Chief Public Prosecutor's Office, Anti-Constitutional Crimes Investigation Bureau from Turkey's Information and Communication Technologies Authority (BTK).

The legal consequences of using flawed CGN logs include extended custody, imprisonment, removal from public service, and other losses with wrongful convictions. The CGN logs provided by the BTK shown in Fig. 4 serves as the primary evidence for a defendant who has been taken into custody in May 2017 and as of July 2020, still on judicial trial with the accusation of being a member of an armed terrorist organization FETO based on the allegation of using ByLock mobile application. He is on trial facing between 7.5 and 15-year prison sentence, like the other FETO membership judicial trials based on allegedly using ByLock (Adal, 2019). Moreover, he was dismissed from civil service. The process of using CGN logs as evidence to put the defendant on trial will be thoroughly explained in Section 5. Below, we explain the technical details about the CGN errors overlooked by legal bodies.

The explanations of the red numbered sections of the CGN shown in Fig. 4 are as below:

Two explicit CGN errors in the example shown in Fig. 4 make the CGN logs flawed.

(1) The missing public IP address and the port number in row 4, where these two variables are mandatory for a session log to be valid in any CGN system (as explained in Section 3.1.).
(2) We propose that if there are significant differences in the public port numbers of a CGN in the same session, there might exist port-jumping errors. For example, when Fig. 4 is analyzed, it can be seen that the minimum public port is 2055 and the maximum public port is 44,076 in the same session (private-IP 10.209.98.165). The port-jumping error can only happen in port range architecture (Boucadair et al., 2009) using CGNs such as deterministic CGN systems. As it is

[1] All translations in this article, including figures and tables, are done by the author.

1	The requesting office and the related file
2	The mobile phone number which was requested to be queried by the BTK for specific destination IP addresses from CGN logs
3	The beginning date of the access query of the destination IP address
4	The end date of the access query of the destination IP address
5	The requested query type, which is CGN logs
6	The results of the internet connection (CGN) communication query
7	Row number
8	Mobile phone number
9	Private IPv4 address of the particular session
10	The private port number of the particular session
11	Public IPv4 address of the particular session
12	The public port number of the particular session
13	Session beginning date
14	Destination IP address
15	Destination port number
16	International Mobile Subscriber Identity (IMSI)
17	International Mobile Equipment Identity (IMEI)
18	Cellular network information (CNI): Consists of call sign ID, operator ID, and physical address of the cellular base station service received
19	Queried mobile phone number, creation date, and the number of pages
20	Investigation year and investigation number of the Office of the Prosecutor

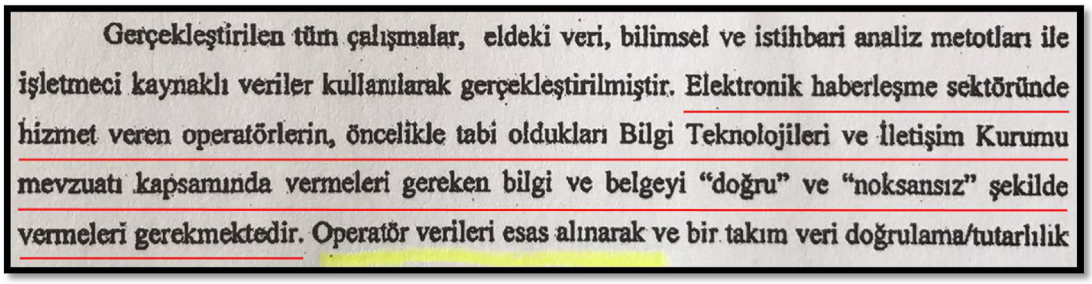

Gerçekleştirilen tüm çalışmalar, eldeki veri, bilimsel ve istihbari analiz metotları ile işletmeci kaynaklı veriler kullanılarak gerçekleştirilmiştir. Elektronik haberleşme sektöründe hizmet veren operatörlerin, öncelikle tabi oldukları Bilgi Teknolojileri ve İletişim Kurumu mevzuatı kapsamında vermeleri gereken bilgi ve belgeyi "doğru" ve "noksansız" şekilde vermeleri gerekmektedir. Operatör verileri esas alınarak ve bir takım veri doğrulama/tutarlılık

Fig. 3. The related part of the MİT's cover letter, which shows how data of ISPs were used (Aktaş, 2018).

explained in Section 4.3.2, it is known that service provider Vodafone, i.e., the service provider of the example given in Fig. 4, uses deterministic NAT44. Therefore, we can conclude that the CGN records example provided in Fig. 4 uses port range architecture. Thus, with a 42,021 public port range, the operator shared a single IP to only a single customer, according to RFC 7422 (Donley et al., 2014). Unfortunately, no more data is provided to the defendants or the courts. Therefore, it is impossible to know port range parameters of that particular session. Nevertheless, with all the data we have, as it is further explained in Section 4.3.2, the educated guess is that port-jumping error might exist in Fig. 4 CGN logs example.

For each internet session, different port ranges can be assigned to the subscribers. Therefore, each session has to be individually analyzed. Furthermore, the assigned port range for that particular internet session has to be known to be sure whether there is a port range error or not. Internet sessions can be inferred from the assigned private IP address and the session-start dates. In some judicial trials, Historical Traffic Search (HTS) records are also requested along with CGN logs. In those cases, the assignment of CGN-log rows to particular internet sessions can be explicitly determined. However, no necessary details were provided to the defendants, such as the assigned port range for that particular session.

4.3.1. Explanation of HTS records

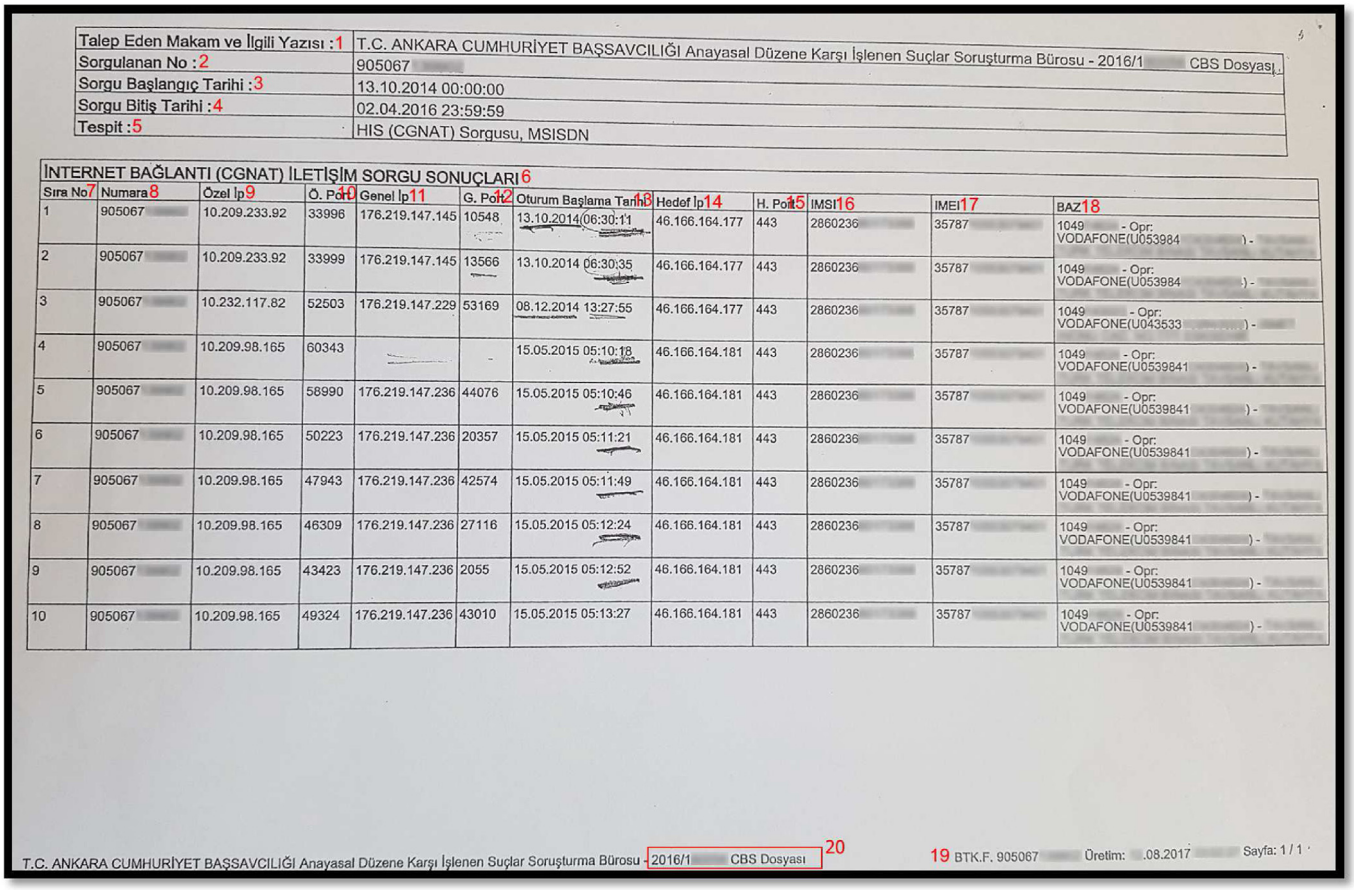
HTS records are mainly composed of two different types of data (Peksayar, 2019): (a) General Packet Radio Services (GPRS)/Wireless Application Protocol (WAP) records, (b) Global System for Mobile Communications (GSM)-related information such as calls, Short Message Service (SMS) messages, and so on.

The relevant part of the HTS file for CGN log analysis is the GPRS records because there has to be an active GPRS session that covers the respective CGN logs with the same private IP address. Irregularities or possible errors discovered in the parent GPRS sessions may reduce dependability or even invalidate the CGN logs used as evidence. The GPRS records contain 12 data fields, as shown in Fig. 5 below.

In Fig. 5, it is possible to see that the destination IP address and the accessed page data columns are empty. This emptiness is expected because internet-connection sessions can hold even millions of connection sessions to thousands of different remote websites. This research has found that only several of the WAP records contained destination IP addresses and accessed pages. When the GPRS records are thoroughly analyzed, it has been found that so many (e.g., 46,782 sessions that overlap with at least six other sessions in Vodafone operator) sessions overlap (a new session starts before the previous one is ended) and sessions are longer than the maximum time interval per GPRS session[2] as shown in Fig. 6 below. Moreover, some (47.28%) GPRS records do not have a CNI value. Therefore, it is inferred that the records having longer session duration than maximum time interval are erroneous.

Moreover, even though no specific details about the version and configuration of the ISP are available, according to the technical

[2] ISPs in Turkey limit GPRS sessions to maximum 7200 s (Peksayar, 2019).



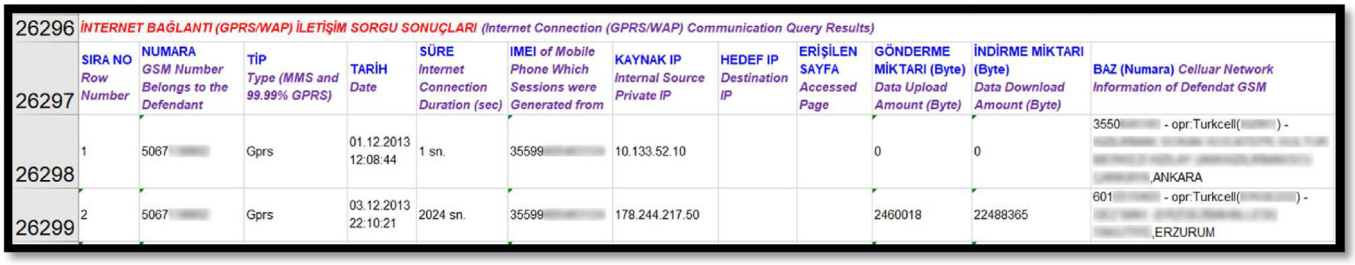
Talep Eden Makam ve İlgili Yazısı :1	T.C. ANKARA CUMHURİYET BAŞSAVCILIĞI Anayasal Düzene Karşı İşlenen Suçlar Soruşturma Bürosu - 2016/1 CBS Dosyası
Sorgulanan No :2	905067
Sorgu Başlangıç Tarihi :3	13.10.2014 00:00:00
Sorgu Bitiş Tarihi :4	02.04.2016 23:59:59
Tespit :5	HIS (CGNAT) Sorgusu, MSISDN

İNTERNET BAĞLANTI (CGNAT) İLETİŞİM SORGU SONUÇLARI 6

Sıra No7	Numara8	Özel Ip9	Ö. Port10	Genel Ip11	G. Port12	Oturum Başlama Tarihi13	Hedef Ip14	H. Port15	IMSI16	IMEI17	BAZ18
1	905067	10.209.233.92	33996	176.219.147.145	10548	13.10.2014 06:30:11	46.166.164.177	443	2860236	35787	1049 - Opr: VODAFONE(U053984) -
2	905067	10.209.233.92	33999	176.219.147.145	13566	13.10.2014 06:30:35	46.166.164.177	443	2860236	35787	1049 - Opr: VODAFONE(U053984) -
3	905067	10.232.117.82	52503	176.219.147.229	53169	08.12.2014 13:27:55	46.166.164.177	443	2860236	35787	1049 - Opr: VODAFONE(U043533) -
4	905067	10.209.98.165	60343			15.05.2015 05:10:18	46.166.164.181	443	2860236	35787	1049 - Opr: VODAFONE(U0539841) -
5	905067	10.209.98.165	58990	176.219.147.236	44076	15.05.2015 05:10:46	46.166.164.181	443	2860236	35787	1049 - Opr: VODAFONE(U0539841) -
6	905067	10.209.98.165	50223	176.219.147.236	20357	15.05.2015 05:11:21	46.166.164.181	443	2860236	35787	1049 - Opr: VODAFONE(U0539841) -
7	905067	10.209.98.165	47943	176.219.147.236	42574	15.05.2015 05:11:49	46.166.164.181	443	2860236	35787	1049 - Opr: VODAFONE(U0539841) -
8	905067	10.209.98.165	46309	176.219.147.236	27116	15.05.2015 05:12:24	46.166.164.181	443	2860236	35787	1049 - Opr: VODAFONE(U0539841) -
9	905067	10.209.98.165	43423	176.219.147.236	2055	15.05.2015 05:12:52	46.166.164.181	443	2860236	35787	1049 - Opr: VODAFONE(U0539841) -
10	905067	10.209.98.165	49324	176.219.147.236	43010	15.05.2015 05:13:27	46.166.164.181	443	2860236	35787	1049 - Opr: VODAFONE(U0539841) -

T.C. ANKARA CUMHURİYET BAŞSAVCILIĞI Anayasal Düzene Karşı İşlenen Suçlar Soruşturma Bürosu - 2016/1 CBS Dosyası 20 19 BTK.F. 905067 Üretim: .08.2017 Sayfa: 1 / 1

Fig. 4. A complete example of CGN records sent from the BTK to the prosecutors in the ByLock cases as evidence (some parts had to be blurred due to the request of the defendant).



İNTERNET BAĞLANTI (GPRS/WAP) İLETİŞİM SORGU SONUÇLARI (Internet Connection (GPRS/WAP) Communication Query Results)

SIRA NO Row Number	NUMARA GSM Number Belongs to the Defendant	TİP Type (MMS and 99.99% GPRS)	TARİH Date	SÜRE Internet Connection Duration (sec)	IMEI of Mobile Phone Which Sessions were Generated from	KAYNAK IP Internal Source Private IP	HEDEF IP Destination IP	ERİŞİLEN SAYFA Accessed Page	GÖNDERME MİKTARI (Byte) Data Upload Amount (Byte)	İNDİRME MİKTARI (Byte) Data Download Amount (Byte)	BAZ (Numara) Celluar Network Information of Defendat GSM
1	5067	Gprs	01.12.2013 12:08:44	1 sn.	35599	10.133.52.10			0	0	3550 - opr:Turkcell( ) - ,ANKARA
2	5067	Gprs	03.12.2013 22:10:21	2024 sn.	35599	178.244.217.50			2460018	22488365	601 - opr:Turkcell( ) - ,ERZURUM

Fig. 5. Column headers and the first two rows of the internet connection (GPRS/WAP) communication query results from the HTS records of the defendant (some parts had to be blurred due to the request of the defendant).

specification documents of 3GPP (P and "Telecommunication m, 2021), these many overlapping GPRS sessions should not exist. According to the documentation, there can be partial data records on some occasions, such as data volume limit, time (duration) limit, management intervention. However, this study has failed to justify these many overlapping sessions and it has concluded that these many overlapping sessions are another indicator of erroneous data records.

A private IP address is assigned to only a single customer for a certain amount of time, and that specific time and private IP address is used to trackback subscribers from logs (Boucadair et al., 2009). Thus, we have scanned the whole HTS file to find both date and private IP address matching GPRS sessions with the CGN provided in Fig. 4. We list all the GPRS sessions (plus one below and above) covering all three Private IP addresses (10.209.233.92, 10.232.117.82, 10.209.98.165) of CGN logs in matching date as shown in Fig. 7 below.

When the GPRS records shown in Fig. 7 are analyzed, there are three irregularities for which this paper has failed to find reasonable answers: (a) The record in row number 51,893 has no upload data traffic and 60 bytes of download data traffic. It is technically impossible to generate an HTTPS web request (row number 3 in Fig. 4) with such an amount of upload and download data traffic; (b) GPRS sessions (rows 40098–40103 and 93999–94004) that have started at the same time with the same duration; (c) CNI in rows 40098–40103 and 93999–94004 that were generated while the defendant was subscribed to another ISP with the same mobile number are missing, whereas the CNI is not missing at the

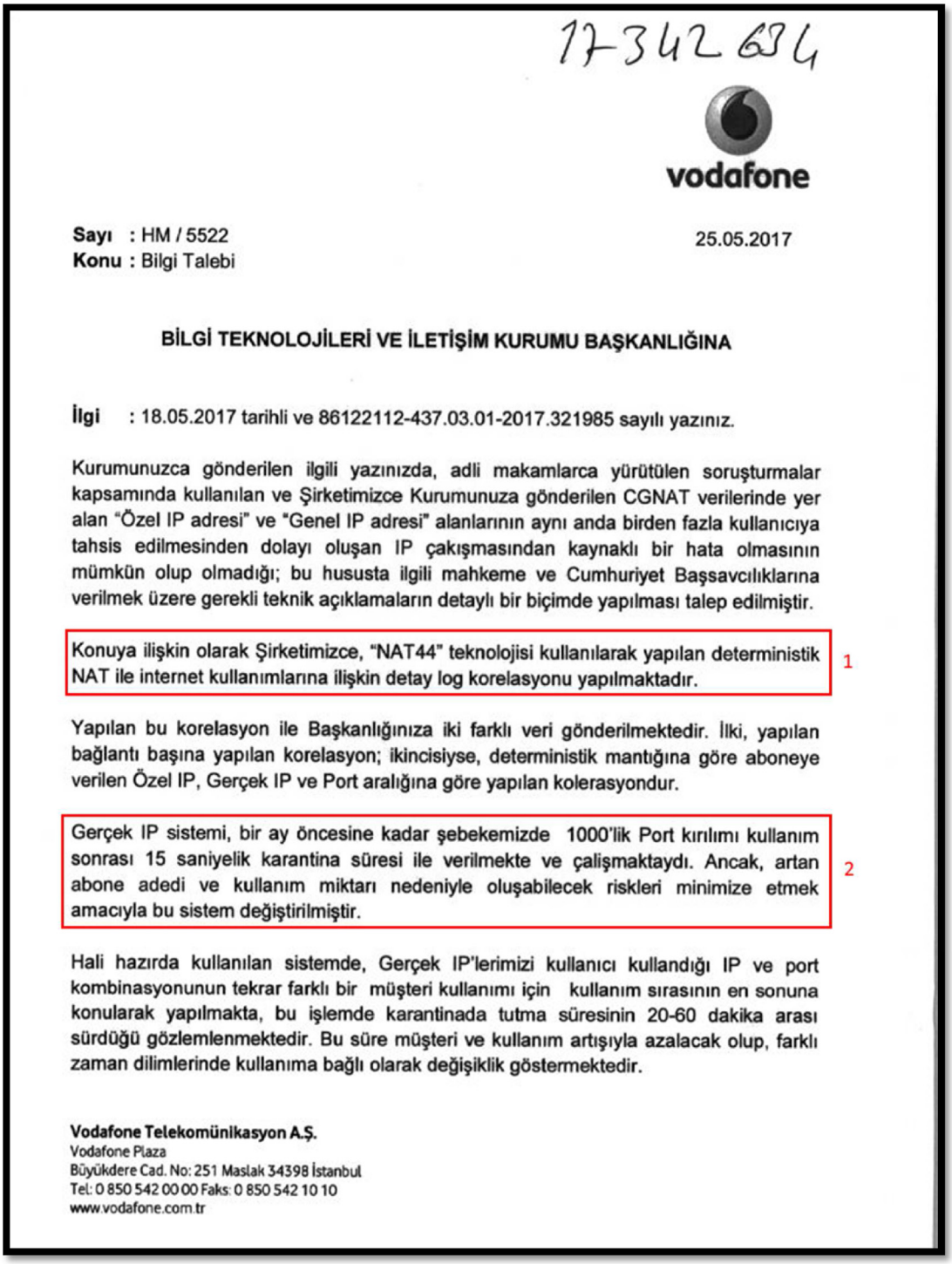
	Row Number	GSM Number Belongs to the Defendant	Type	Date	Internet Connection Duration (sec)	IMEI	Source Private IP	Destination IP	Accessed Page	Data Upload Amount (Byte)	Data Download Amount (Byte)	Cellular Network Information of Defendat GSM
26708												
26709	411	5067	Gprs	07.02.2014 19:31:33	41890 sn.	35599	10.59.5.47			37010	88338	4 77 - opr:Avea
26710	412	5067	Gprs	07.02.2014 19:31:33	41890 sn.	35599	10.59.5.47			7488366	92183033	4 77 - opr:Avea
26711	413	5067	Gprs	07.02.2014 19:31:33	41890 sn.	35599	10.59.5.47			736403	4964604	4 77 - opr:Avea
26712	414	5067	Gprs	07.02.2014 19:31:34	41889 sn.	35599	10.59.5.47			7817	7826	4 77 - opr:Avea
26713	415	5067	Gprs	07.02.2014 19:31:38	41885 sn.	35599	10.59.5.47			2654	4094	4 77 - opr:Avea
26714	416	5067	Gprs	07.02.2014 22:24:43	31500 sn.	35599	10.59.5.47			6263	11494	4 77 - opr:Avea
26715	417	5067	Gprs	08.02.2014 06:19:20	3023 sn.	35599	10.59.5.47			5697	26891	4 77 - opr:Avea
26716	418	5067	Gprs	08.02.2014 18:21:40	46800 sn.	35599	10.51.115.8			32946	80830	4 58 - opr:Avea
26717	419	5067	Gprs	08.02.2014 18:21:40	46800 sn.	35599	10.51.115.8			1094877	28622907	4 58 - opr:Avea
26718	420	5067	Gprs	08.02.2014 18:21:40	46800 sn.	35599	10.51.115.8			3142788	35936259	4 58 - opr:Avea
26719	421	5067	Gprs	08.02.2014 18:21:41	46800 sn.	35599	10.51.115.8			6951	7360	4 58 - opr:Avea
26720	422	5067	Gprs	08.02.2014 18:21:42	46800 sn.	35599	10.51.115.8			6165	6390	4 58 - opr:Avea
26721	423	5067	Gprs	08.02.2014 18:22:37	46800 sn.	35599	10.51.115.8			848	9314	4 58 - opr:Avea
26722	424	5067	Gprs	08.02.2014 19:15:47	47632 sn.	35599	10.51.115.8			6017	0	4 58 - opr:Avea
26723	425	5067	Gprs	08.02.2014 21:49:16	38423 sn.	35599	10.51.115.8			50557	637730	4 58 - opr:Avea
26724	426	5067	Gprs	09.02.2014 07:21:40	4079 sn.	35599	10.51.115.8			10350	25410	4 58 - opr:Avea
26725	427	5067	Gprs	09.02.2014 07:21:40	4079 sn.	35599	10.51.115.8			808951	5188298	4 58 - opr:Avea
26726	428	5067	Gprs	09.02.2014 07:21:40	4079 sn.	35599	10.51.115.8			4895157	267445972	4 58 - opr:Avea
26727	429	5067	Gprs	09.02.2014 07:21:41	4078 sn.	35599	10.51.115.8			1096	1308	4 58 - opr:Avea
26728	430	5067	Gprs	09.02.2014 07:21:42	4077 sn.	35599	10.51.115.8			0	0	4 58 - opr:Avea
26729	431	5067	Gprs	09.02.2014 07:22:37	4022 sn.	35599	10.51.115.8			0	0	4 58 - opr:Avea
26730	432	5067	Gprs	09.02.2014 16:29:55	45110 sn.	35599	10.48.177.222			23543	56804	4 59 - opr:Avea
26731	433	5067	Gprs	09.02.2014 16:29:55	45110 sn.	35599	10.48.177.222			821807	21375773	4 59 - opr:Avea

Fig. 6. An example of colliding GPRS and 7200 s exceeding GPRS session records from the HTS records of the defendant (some parts had to be blurred due to the request of the defendant).

Row Number	GSM Number Belongs to the Defendant	Type	Date	Internet Connection Duration (sec)	IMEI	Source Private IP	Destination IP	Accessed Page	Data Upload Amount (Byte)	Data Download Amount (Byte)	Cellular Network Information of Defendat GSM
40097	5067	Gprs	13.10.2014 05:55:18	359 sn.	35787	10.209.40.247			0	0	0-Kayit Bulunamadi (No Record Found)
40098	5067	Gprs	13.10.2014 06:30:03	1882 sn.	35787	10.209.233.92			104726	840262	0-Kayit Bulunamadi
40099	5067	Gprs	13.10.2014 06:30:03	1882 sn.	35787	10.209.233.92			67955	876557	0-Kayit Bulunamadi
40100	5067	Gprs	13.10.2014 06:30:03	1882 sn.	35787	10.209.233.92			0	0	0-Kayit Bulunamadi
40101	5067	Gprs	13.10.2014 06:30:03	1882 sn.	35787	10.209.233.92			10747	19120	0-Kayit Bulunamadi
40102	5067	Gprs	13.10.2014 06:30:03	1882 sn.	35787	10.209.233.92			39173	102280	0-Kayit Bulunamadi
40103	5067	Gprs	13.10.2014 06:30:03	1882 sn.	35787	10.209.233.92			82244	655004	0-Kayit Bulunamadi
40104	5067	Gprs	13.10.2014 11:32:54	1210 sn.	35787	10.211.180.74			30026	45885	0-Kayit Bulunamadi
51892	5067	Gprs	08.12.2014 13:16:39	385 sn.	35787	5.229.149.79			0	125	1049 - opr:Vodafone(UA43
51893	5067	Gprs	08.12.2014 13:24:24	3599 sn.	35787	10.232.117.82			0	60	1049 - opr:Vodafone(UA43
51894	5067	Gprs	08.12.2014 14:32:55	37 sn.	35787	176.218.189.116			28726	58814	5260 - opr:Vodafone(GB43
93998	5067	Gprs	15.05.2015 04:13:28	360 sn.	35787	10.209.108.156			0	0	0-Kayit Bulunamadi
93999	5067	Gprs	15.05.2015 05:10:02	1219 sn.	35787	10.209.98.165			33562	262471	0-Kayit Bulunamadi
94000	5067	Gprs	15.05.2015 05:10:02	1219 sn.	35787	10.209.98.165			7142	13858	0-Kayit Bulunamadi
94001	5067	Gprs	15.05.2015 05:10:02	1219 sn.	35787	10.209.98.165			78711	865927	0-Kayit Bulunamadi
94002	5067	Gprs	15.05.2015 05:10:02	1219 sn.	35787	10.209.98.165			73668	772130	0-Kayit Bulunamadi
94003	5067	Gprs	15.05.2015 05:10:02	1219 sn.	35787	10.209.98.165			55108	889316	0-Kayit Bulunamadi
94004	5067	Gprs	15.05.2015 05:10:03	1218 sn.	35787	10.209.98.165			0	0	0-Kayit Bulunamadi
94005	5067	Gprs	15.05.2015 06:00:22	373 sn.	35787	10.208.102.230			1816	2890	0-Kayit Bulunamadi

Fig. 7. GPRS sessions that cover the CGN logs shown in Fig. 4 from the HTS records of the defendant (some parts had to be blurred due to the request of the defendant).

respective CGN logs shown in Fig. 4. The CNI difference between different ISPs is another proof that the source of CNI is the same for CGN logs and GPRS logs. From data irregularities shown in Figs. 6 and 7, one can logically deduct that not only the logs of implemented CGN-systems but also the logs of data services are error-prone as well.



17342684

vodafone

Sayı : HM / 5522
Konu : Bilgi Talebi

25.05.2017

BİLGİ TEKNOLOJİLERİ VE İLETİŞİM KURUMU BAŞKANLIĞINA

İlgi : 18.05.2017 tarihli ve 86122112-437.03.01-2017.321985 sayılı yazınız.

Kurumunuzca gönderilen ilgili yazınızda, adli makamlarca yürütülen soruşturmalar kapsamında kullanılan ve Şirketimizce Kurumunuza gönderilen CGNAT verilerinde yer alan "Özel IP adresi" ve "Genel IP adresi" alanlarının aynı anda birden fazla kullanıcıya tahsis edilmesinden dolayı oluşan IP çakışmasından kaynaklı bir hata olmasının mümkün olup olmadığı; bu hususta ilgili mahkeme ve Cumhuriyet Başsavcılıklarına verilmek üzere gerekli teknik açıklamaların detaylı bir biçimde yapılması talep edilmiştir.

Konuya ilişkin olarak Şirketimizce, "NAT44" teknolojisi kullanılarak yapılan deterministik NAT ile internet kullanımlarına ilişkin detay log korelasyonu yapılmaktadır. 1

Yapılan bu korelasyon ile Başkanlığınıza iki farklı veri gönderilmektedir. İlki, yapılan bağlantı başına yapılan korelasyon; ikincisiyse, deterministik mantığına göre aboneye verilen Özel IP, Gerçek IP ve Port aralığına göre yapılan kolerasyondur.

Gerçek IP sistemi, bir ay öncesine kadar şebekemizde 1000'lik Port kırılımı kullanım sonrası 15 saniyelik karantina süresi ile verilmekte ve çalişmaktaydı. Ancak, artan abone adedi ve kullanım miktarı nedeniyle oluşabilecek riskleri minimize etmek amacıyla bu sistem değiştirilmiştir. 2

Hali hazırda kullanılan sistemde, Gerçek IP'lerimizi kullanıcı kullandığı IP ve port kombinasyonunun tekrar farklı bir müşteri kullanımı için kullanım sırasının en sonuna konularak yapılmakta, bu işlemde karantinada tutma süresinin 20-60 dakika arası sürdüğü gözlemlenmektedir. Bu süre müşteri ve kullanım artışıyla azalacak olup, farklı zaman dilimlerinde kullanıma bağlı olarak değişiklik göstermektedir.

Vodafone Telekomünikasyon A.Ş.
Vodafone Plaza
Büyükdere Cad. No: 251 Maslak 34398 İstanbul
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www.vodafone.com.tr

Fig. 8. Vodafone's official reply to the BTK request regarding the ByLock prosecutions data that are sent to the law enforcement, HM/5522 (Peksayar, 2018).

4.3.2. Port-jumping error and its possible consequences

In a dynamic-port allocating CGN system, public port-jumping would not be an error. So, the question is how to determine the port-jumping error in the provided CGN logs in Fig. 4? (a) The ISP shown in column BAZ (number 18) is the ISP called Vodafone Turkey. (b) Vodafone has answered an official request (Id: HM/5522) from the BTK and stated that they use deterministic NAT44 technology, as shown in Fig. 4 below. (c) Vodafone Turkey has 1,655,000 public IPv4 addresses reserved while they have 20 million customers (Dulkadir et al., 2018). Thus, on average, 12 subscribers per public IPv4 address have to be assigned. However, the port range in rows 5–10 is 42,021, which would require a single IPv4 address to be fully dedicated to a customer. When considering all factors, in this case, port-jumping errors may exist in the CGN logs shown in Fig. 4. The port-jumping error has never been discovered or mentioned in the literature before, to the best of our knowledge.

Vodafone's answer shown in Fig. 8 above clearly states that they

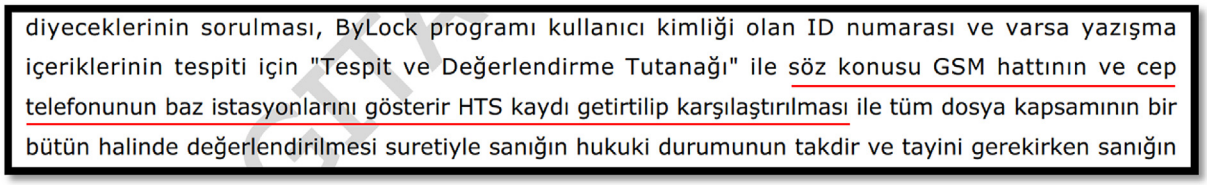

diyeceklerinin sorulması, ByLock programı kullanıcı kimliği olan ID numarası ve varsa yazışma içeriklerinin tespiti için "Tespit ve Değerlendirme Tutanağı" ile söz konusu GSM hattının ve cep telefonunun baz istasyonlarını gösterir HTS kaydı getirtilip karşılaştırılması ile tüm dosya kapsamının bir bütün halinde değerlendirilmesi suretiyle sanığın hukuki durumunun takdir ve tayini gerekirken sanığın

Fig. 9. Related part of the case 2017/3652 (E), 2018/262 (K) which can be searched at the official website https://karararama.yargitay.gov.tr/YargitayBilgiBankasiIstemciWeb/.

used deterministic NAT and shared 1000 ports per customer until 2017. The CGN logs provided in Fig. 4 are from 2015. The translation of paragraph-1 and -2 in Fig. 8 is as follows:

(1) "Regarding the subject, our Company performs detail-log correlations regarding internet usage with deterministic NAT that is made by using NAT44 technology."
(2) "The real IP system used to be provided and operate in our network with 15-s quarantine duration after 1000 port breakage usage until a month ago. However, this system has been changed to minimize the risk that might occur due to the increasing number of subscribers and usage amount."

A port-jumping error may lead to incorrect subscriber identification from the deterministic CGN logs because the port range is used to trackback individual subscribers in deterministic CGNs. Moreover, the existence of a port-jumping error means that either the configuration of the CGN was incorrect or the logging system was flawed. In both cases, the CGN logs become undependable and flawed.

4.4. The possible causes of flaws in CGN logs

Since setups of CGN systems deployed by ISPs are hidden from the public, it is most unlikely to determine what can cause flaws in CGN logs, as explained in section 4.3. However, the educated guesses about the possible causes of potential flaws in CGN logs can be listed as below:

- As mentioned in the OFCOM report (Aitken, 2013), to provide meaningful and correct information to law enforcement, ISPs must keep enormous log data that will not bring any direct monetary gain. Moreover, incorrectly keeping CGN logs does not negatively affect their customers' experience. Because of these reasons, ISPs may not pay proper attention needed for CGN logging, especially if there are no strict regulations and fines (e.g., we have failed to find any regulations for ISPs in Turkey).
- Software bugs can cause flaws in CGN logging, as happened in the case of the F5 Networks (2018a).
- Incorrect parameter setups can cause flaws in CGN logs. Usually, the used CGN system should not allow an operator to make an erroneous configuration. However, as in the F5 Networks case (Networks, 2018b), a specific configuration may cause logs to be flawed.
- Typographical errors made by authorized staff (Drury, 2017).

4.5. The process that led the Turkish court of caseation to develop an erroneous approach to verify the correctness of CGN logs

This section explains the significance of the technical findings above from a legal standpoint in detail with examples.

The 16th Criminal Chamber of the Court of Cassation, i.e., the higher court that handles terrorism cases and delivers the final decision in Turkey, has developed jurisprudence to verify the integrity of CGN logs presented to the criminal courts by the BTK. For example, this verification approach is expressed in case 2017/3652 (E), 2018/262 (K), as shown in Fig. 9 below.

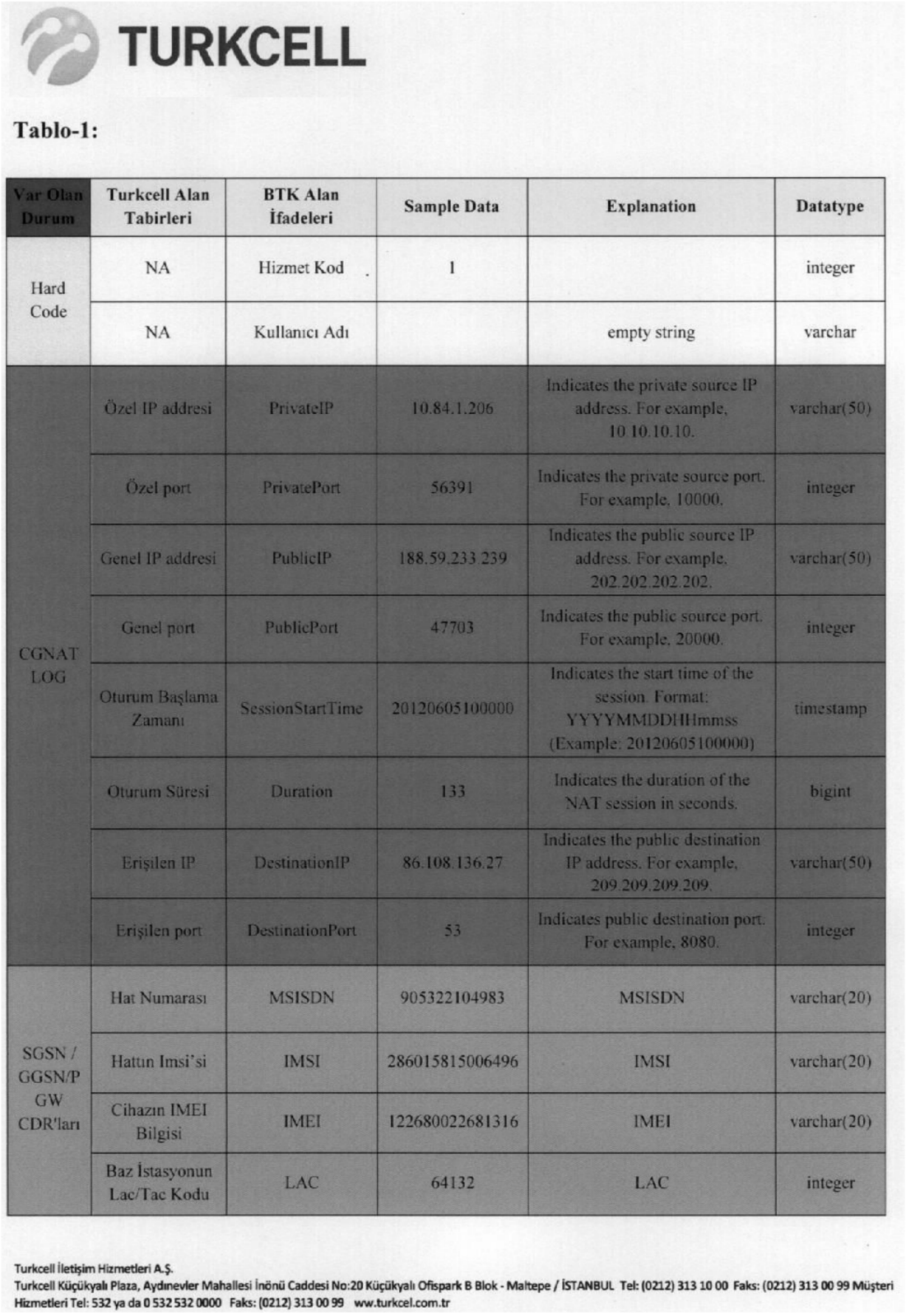
The underlined sentence shown in Fig. 9 can be translated as "HTS records of the GSM line and the mobile phone in question that show cellular base station to be brought on and compared". The court suggests the comparison of the CNI provided in the CGN logs file, as shown in Fig. 4 (red number 18), and the HTS logs file, as shown in Fig. 7. This verification approach is also mentioned as credible evidence in a news report of the ByLock judicial case imprisonment to 6 years and 3 months (Özkaya, 2019). However, this approach is fundamentally incorrect. Because, in CGN logs, the cellular station information is not stored, and this is verified by Turkcell (Peksayar, 2018), which is one of the major mobile ISPs in Turkey. Turkcell has officially answered (No: LC/NS/5274884) a request of the BTK and displayed which variables are kept as logs for Customer Data Record (CDR) and CGN sessions, as shown in Fig. 10 below. When the data provided by Turkcell in Fig. 10 is analyzed, it can be seen that information related to the cellular station is only stored at the CDR logs of Serving (General Packet Radio Service) Support Node (SGSN), Gateway GPRS Support Node (GGSN) and Packet Data Network Gateway (PGW). Moreover, as shown in Fig. 11 below, Turkcell stated that the CGN logs sent to them for analysis are composed by mapping two independent sources, which can be listed as (a) The CDR logs of SGSN and GGSN; (b) A10 infrastructures which form CGN.

Criminal courts have used this fundamentally flawed verification approach in the judicial ByLock cases against the incorrect CGN logs argument of defendants.

The answer of Turkcell (No: LC/NS/5274884) to the BTK also includes detailed information about how CGN and CDR logs were mapped together, as shown in Fig. 12 below. In addition, Turkcell states that all of the necessary conditions stated below have to be ensured for logs to be determined as valid:

- The private IP addresses of SGSN/GGSN (named as GPRS in HTS file) CDR and CGN logs have to match
- The session start date of SGSN/GGSN CDR logs has to be equal to or smaller than the session start date of CGN logs
- The session end date of SGSN/GGSN CDR logs has to be bigger than the session start date of CGN logs

Moreover, if there are two or more same private IP addresses during the same session date, such logs are not delivered to the BTK from Turkcell. Assuming that all of the operators have followed these approaches, these rules explain why this study has failed to find any CGN logs that are not covered by any GPRS session in the HTS records that have been analyzed. However, these rules still do not explain the irregularities in records of GPRS sessions and the errors in the CGN logs that have been determined.



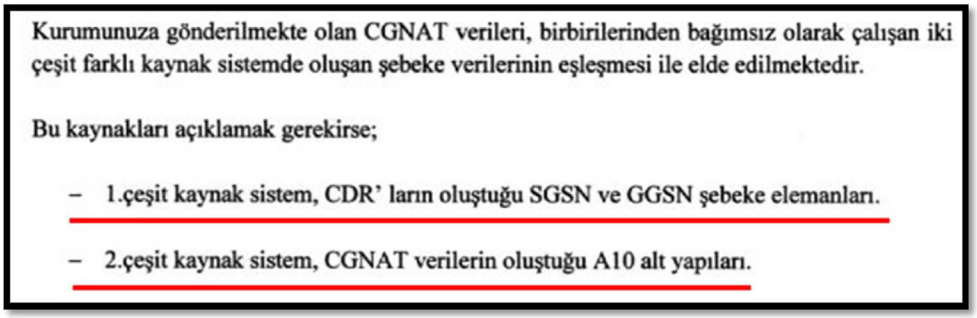
TURKCELL

Tablo-1:

Var Olan Durum	Turkcell Alan Tabirleri	BTK Alan İfadeleri	Sample Data	Explanation	Datatype
Hard Code	NA	Hizmet Kod	1		integer
	NA	Kullanıcı Adı		empty string	varchar
CGNAT LOG	Özel IP addresi	PrivateIP	10.84.1.206	Indicates the private source IP address. For example, 10.10.10.10.	varchar(50)
	Özel port	PrivatePort	56391	Indicates the private source port. For example, 10000.	integer
	Genel IP addresi	PublicIP	188.59.233.239	Indicates the public source IP address. For example, 202.202.202.202.	varchar(50)
	Genel port	PublicPort	47703	Indicates the public source port. For example, 20000.	integer
	Oturum Başlama Zamanı	SessionStartTime	20120605100000	Indicates the start time of the session. Format: YYYYMMDDHHmmss (Example: 20120605100000)	timestamp
	Oturum Süresi	Duration	133	Indicates the duration of the NAT session in seconds.	bigint
	Erişilen IP	DestinationIP	86.108.136.27	Indicates the public destination IP address. For example, 209.209.209.209.	varchar(50)
	Erişilen port	DestinationPort	53	Indicates public destination port. For example, 8080.	integer
SGSN / GGSN/P GW CDR'ları	Hat Numarası	MSISDN	905322104983	MSISDN	varchar(20)
	Hattın Imsi'si	IMSI	286015815006496	IMSI	varchar(20)
	Cihazın IMEI Bilgisi	IMEI	122680022681316	IMEI	varchar(20)
	Baz İstasyonun Lac/Tac Kodu	LAC	64132	LAC	integer

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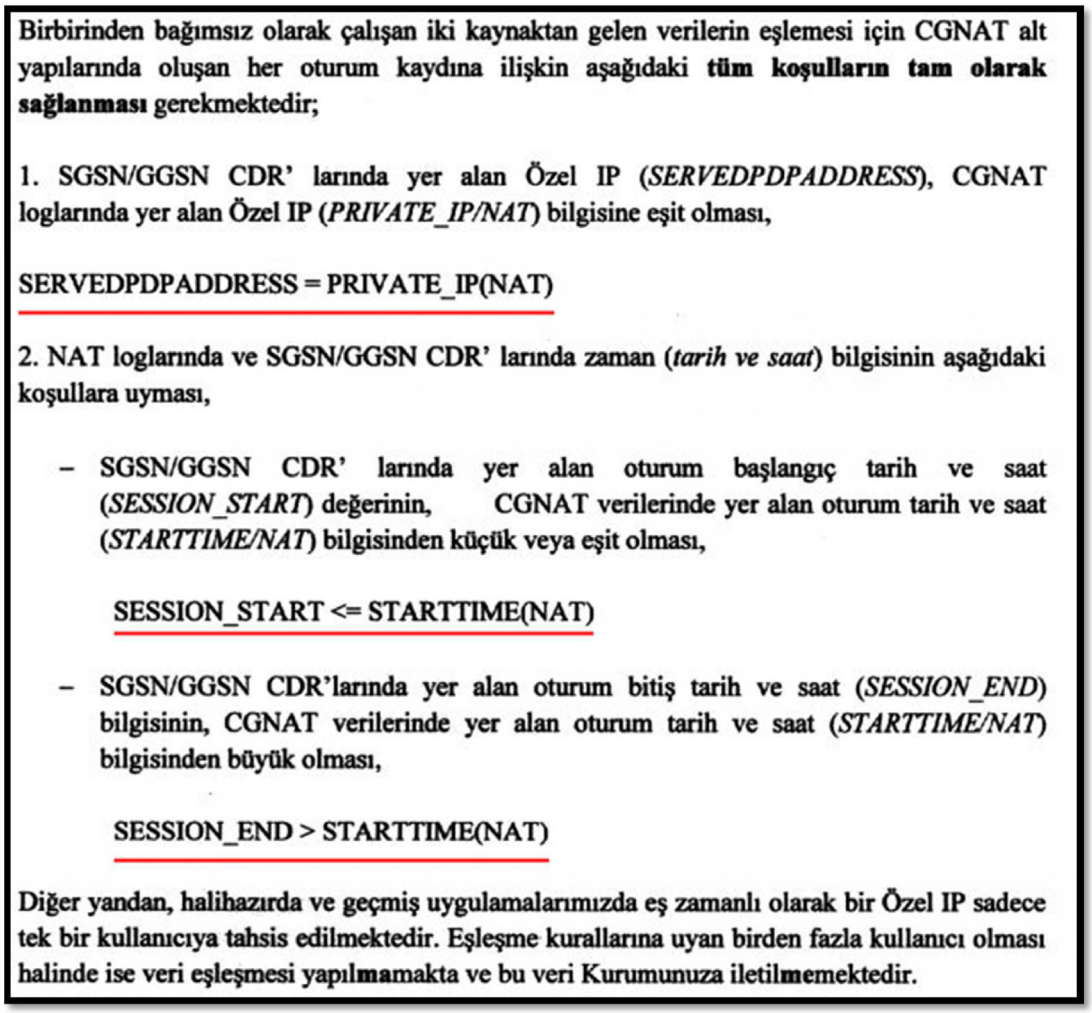
Fig. 10. Detailed explanation of how Turkcell keeps CGN and CDR logs (Peksayar, 2018).


Kurumunuza gönderilmekte olan CGNAT verileri, birbirilerinden bağımsız olarak çalışan iki çeşit farklı kaynak sistemde oluşan şebeke verilerinin eşleşmesi ile elde edilmektedir.

Bu kaynakları açıklamak gerekirse;

- 1.çeşit kaynak sistem, CDR' ların oluştuğu SGSN ve GGSN şebeke elemanları.
- 2.çeşit kaynak sistem, CGNAT verilerin oluştuğu A10 alt yapıları.

Fig. 11. The explanation of how the CGN logs sent from the Turkcell are composed (Peksayar, 2018).


Birbirinden bağımsız olarak çalışan iki kaynaktan gelen verilerin eşlemesi için CGNAT alt yapılarında oluşan her oturum kaydına ilişkin aşağıdaki **tüm koşulların tam olarak sağlanması** gerekmektedir;

1. SGSN/GGSN CDR' larında yer alan Özel IP (*SERVEDPDPADDRESS*), CGNAT loglarında yer alan Özel IP (*PRIVATE_IP/NAT*) bilgisine eşit olması,

SERVEDPDPADDRESS = PRIVATE_IP(NAT)

2. NAT loglarında ve SGSN/GGSN CDR' larında zaman (*tarih ve saat*) bilgisinin aşağıdaki koşullara uyması,

- SGSN/GGSN CDR' larında yer alan oturum başlangıç tarih ve saat (*SESSION_START*) değerinin, CGNAT verilerinde yer alan oturum tarih ve saat (*STARTTIME/NAT*) bilgisinden küçük veya eşit olması,

  SESSION_START <= STARTTIME(NAT)

- SGSN/GGSN CDR'larında yer alan oturum bitiş tarih ve saat (*SESSION_END*) bilgisinin, CGNAT verilerinde yer alan oturum tarih ve saat (*STARTTIME/NAT*) bilgisinden büyük olması,

  SESSION_END > STARTTIME(NAT)

Diğer yandan, halihazırda ve geçmiş uygulamalarımızda eş zamanlı olarak bir Özel IP sadece tek bir kullanıcıya tahsis edilmektedir. Eşleşme kurallarına uyan birden fazla kullanıcı olması halinde ise veri eşleşmesi yapılmamakta ve bu veri Kurumunuza iletilmemektedir.

Fig. 12. Detailed explanation of how Turkcell maps CGN and CDR logs from the answer (No: LC/NS/5274884) (Peksayar, 2018).

4.6. Subscriber tracking-based incorrect judicial cases outside of Turkey

Tracking crime suspects from IP addresses is an internationally used methodology. However, as we explained, when the subscriber is behind a NAT, it becomes challenging to determine the true owner of the IP address. Even truly detecting the subscriber of the IP address still does not guarantee that the subscriber has committed the crime. This study could not find any reverse-tracking based on any judicial cases other than those in Turkey. However, there are cases where IP address and ISP-based data led to false arrests and judicial trials.

Interception of Communications Commissioner of the UK has raised severe concerns about IP address resolutions in their 2016 report ("Report of the Intercepti, 2017). According to the report, many of the serious errors were caused when resolving IP addresses. They discovered errors in logs of Communications Service Providers (CSPs), which could lead to the misidentification of subscribers.

Four solid cases of how IP address resolution harmed innocent

subscribers in the United States (US) were extensively discussed in (Mackey et al., 2016). In the first case, the police forces used physical addresses provided by third parties to raid an unrelated farm based on the given IP address. In the second case, the police forces raided the home of two activists who allowed their network to be one of The Onion Routing (Tor)[3] exit relay. In the third case, the police forces raided and arrested a man for child pornography. However, it later turned out that the IP address owner was not the perpetrator but the neighbor using his open wireless network. The fourth given example case is about adult film copyright infringement. In the fourth case, the court ruled that IP addresses cannot be counted as individuals. It is worth noting that in all four cases, no reverse-tracking method was used. There was a crime, and there was an IP address of the crime. However, tracking the subscriber of the IP address led to incorrect individuals.

Denmark's authority's decision to review over 10,000 court verdicts due to two critical errors discovered in their cellular networks system is another excellent example of why logs of ISPs should be cautiously used as evidence (Sorensen, 2019). They used the cellular location as evidence in judicial cases to sentence crime suspects. Keeping logs of location information is much simpler and requires lesser resources than CGN-logs. Because location logs would fundamentally require fewer parameters when compared to CGN-logs such as timestamp, subscriber ID, and connected station ID. However, even at location logs, errors led to the arrest of at least 32 innocent people in Denmark's case (Henley, 2019).

5. Possible severe consequences of reverse-tracking from CGN logs

As explained in Section 4.1, when cybercrime is committed, the public IP address and the port number of the connection are provided to the ISPs to identify the subscriber of the connection. However, reverse-tracking from CGN logs follows a different approach. At this methodology, all subscribers who tried to connect to a particular destination IP address during a specific period are identified as suspects. This approach is fundamentally erroneous for several reasons, which will be explained using actual ByLock cases.

To identify ByLock users, the MİT and the BTK have worked together, as explained by the MİT's official document (Dulkadir et al., 2018). The MİT states that whoever has accessed the application in at least three different days has been put on the list of ByLock users. With this methodology, they have identified 102,192 different people with 123,115 different mobile phone numbers and 6,748 fixed-line subscriptions. Moreover, they state that the actual owner of the mobile phone number may vary because someone else may be the actual user of the mobile phone number, and more than one individual may have connected from fixed-line subscriptions. When the provided numbers are analyzed, there are two eye-catching issues: (a) How can the mobile phone identification count be 18.2 times that of fixed-line subscriptions? The 18.2 times ratio is against the normal flow of life when internet subscribers in Turkey are considered. People usually have fixed-line subscriptions along with mobile subscriptions since the mobile subscription is still inferior to the fixed-line subscription in Turkey (e.g., 4 GB data traffic in mobile subscription starts around 5$; while unlimited data traffic fixed-line subscription starts around 12$ per month). Moreover, the date of reverse-tracking CGN-logs scan to determine the list of ByLock users was between April 2014 and April 2016, when the mobile traffic quotes were even lower than today. So, considering this gap, the educated guess of this research is that many fixed-line operators did not save any useful CGN-logs; (b) They explicitly state that the subscribers must have accessed the application for at least three days to be put on the list. It is not known why subscribers who accessed the application for one- or-two day were excluded from the list. However, the reason behind it may be that the number of suspects would be much higher if those subscribers were also included. According to the official announcement made by the Turkish Deputy Minister of Interior, 79,337 people were subjected to judicial proceedings as of March 2019 (Bulur and Dirim, 2019). Furthermore, he has said that 95,310 people were identified to be a user of ByLock so far.

How exactly these people were identified is explained in more detail in the MİT's ByLock Application Technical Report (MİT, 2017). According to the MİT's report, nine different IP addresses were used by the server of the ByLock application. So, the BTK has reverse scanned these IP addresses and determined who has ever made connections to one of these IP addresses from the destination IP address of CGN logs. We could not find any information related to the beginning and the end date of this reverse scan. However, according to a well-known forensics expert in Turkey, on 14 August 2014, the ByLock server was moved from the US to Lithuania, and the previously used IP addresses in the US were not included in the reverse scan (Beşikçi, 2017a). Hence, it can be inferred that the beginning date of CGN logs reverse scanning was at 14 August 2014. It is estimated that the end date of the reverse scan is around April 2016. According to the ByLock statistics provided by AppBrain (2019), ByLock was put on the Google Market on 11 April 2014 and removed from the Google Market on 3 April 2016. Moreover, it has been installed more than 100,000 times as of 19 January 2015. During the research, it has been observed that the earliest date of the ByLock CGN log sent to the courts was August 2014.

When criminal courts or prosecutors request CGN logs as evidence from the BTK, the beginning date of the request is the first date of the CGN log that has the destination IP address of one of the ByLock server IP addresses, as seen in Fig. 4. In the example given in Fig. 4, the beginning date of the query requested from the BTK is 13 October 2014, and the end date is 2 April 2016. When the logs are analyzed, the beginning date of sessions in the first row is 13 October 2014, 6:30 AM. The date of the first ByLock log is written at the ByLock Detection Report (BDR), which is sent to all prosecutors to start a judicial trial. So, based on the ByLock-users list, a BDR is generated for all suspects and sent to the prosecutors to start a judicial prosecution. Once the prosecution begins, more detailed CGN logs from the BTK are requested as evidence. In 2018, this procedure was slightly modified due to the “Morbeyin event” which will be explained in section 5.1.

There are four critical issues with the reverse-tracking approach: (a) To reverse-track from CGN logs, the ISP has to keep the destination IP address and the port number of each session. Keeping destination IP address logs is seen as a privacy issue (Perreault et al., 2013). (b) Since a massive number of sessions usually gets scanned, the error rate of the logs has to be correctly estimated to evaluate the evidential value of logs accurately. Any error in CGN logs may cause unrelated individuals to be put on the list. (c) At present, many applications share the same public IP address with the same public port number. This invalidates the argument that the users of the software can be determined from its server IP address. For instance, the MİT and the BTK used this approach to identify ByLock users. (d) Any software can redirect anyone to any IP address. For example, a news website can display an image that links to website X. Thus, the user's browser/app would try to access and download that image from the IP address of website X. Accessing the website X would get logged in CGN logs even if that image does not exist on website X. This scenario (*d*) is not depended on any form of CGN logging. It can be either direct

[3] Tor: Overview, https://2019.www.torproject.org/about/overview.html.en.

logging of IPv6, deterministic CGN, or standard CGN. Furthermore, the CGN logs provided by the BTK have only the beginning of the session request. Thus, no information regarding the duration, status, and download/upload data traffic of individual sessions in CGN logs is provided by the BTK to the criminal courts. Therefore, it is impossible to evaluate what each session means correctly in CGN logs. Consequently, there is no way to differentiate and determine the source of the connection request properly.

5.1. "Morbeyin" event that led thousands of people to be released from prison and to get an acquittal

The 16th Criminal Chamber of the Court of Cassation made the first ruling about ByLock on 14 July 2017 (2017/1443 E, 2017/4758 K) and accepted ByLock usage as definitive evidence of FETO membership (Medeniyeti, 2018). Since membership to an armed terrorist organization is a catalog crime, ByLock defendants were usually jailed pending trial. Moreover, since the CNI match of GPRS and CGN logs was seen enough to verify ByLock usage by the Turkish Court of Cassation, many criminal courts were sentencing people as members of an armed terrorist organization based on CGN logs. However, this procedure has changed with the so-called "Morbeyin event".

Lawyer Aktaş has obtained the screenshot, which led to the Morbeyin event displayed in Fig. 13 below. This screenshot was taken from a formal technical analysis of a mobile phone which belongs to a ByLock case defendant. Then, he has sent this obtained information to the digital forensic expert Peksayar to prepare a technical report (Peksayar, 2017). The screenshot was taken from a digital forensic analysis report of a mobile phone officially requested by the ACPPO in a ByLock case. Therefore, it was an undeniable evidence. The screenshot in Fig. 13 shows information found in Android Web View buffer memory (Peksayar, 2017). Web View is used to display web pages inside an Android application. When the found temporary file is analyzed, it is discovered that an iframe loads the https://bylock.net inside the application named Freezy. The Freezy app was an application produced by a developer team called MorBeyin (Peksayar, 2017). Once Aktaş received the report, he applied to the ACPPO for this application to be investigated (Aktaş) and then announced this application to the public. Until this finding was documented, the list of ByLock users composed by reverse-scanning of CGN logs was accepted as flawless. Therefore, solely based on CGN logs, people were being sentenced as members of armed terrorist organization. Although something like the behavior of the Freezy app was always a possibility, authorities never accepted it until this finding. Thus, the finding of the Freezy app was a turning point in the mass prosecutions.

Later on, ACPPO announced that they had identified 11,480 GSM subscribers who had been redirected to ByLock app without their permission (Akcay, 2017). However, they did not provide any details about how they have determined these 11,480 GSM numbers. Besides, Ankara Chief Public Prosecutor stated that around 1000 imprisoned alleged ByLock users would be released. However, the number of people among those 11,480 subscribers who were already being prosecuted, sentenced, or imprisoned previously was not stated. So, the list of ByLock users was updated once again. Even though the authorities claimed once again that the latest list was flawless (all users were 100% ByLock users according to the authorities), we believe that it was not the case because any application-user list is composed by the reverse-tracking of IP logs methodology is fundamentally vulnerable to errors. As this paper has explained, even if CGN logs are accurate enough to be evidential, the determined user might still not have used the

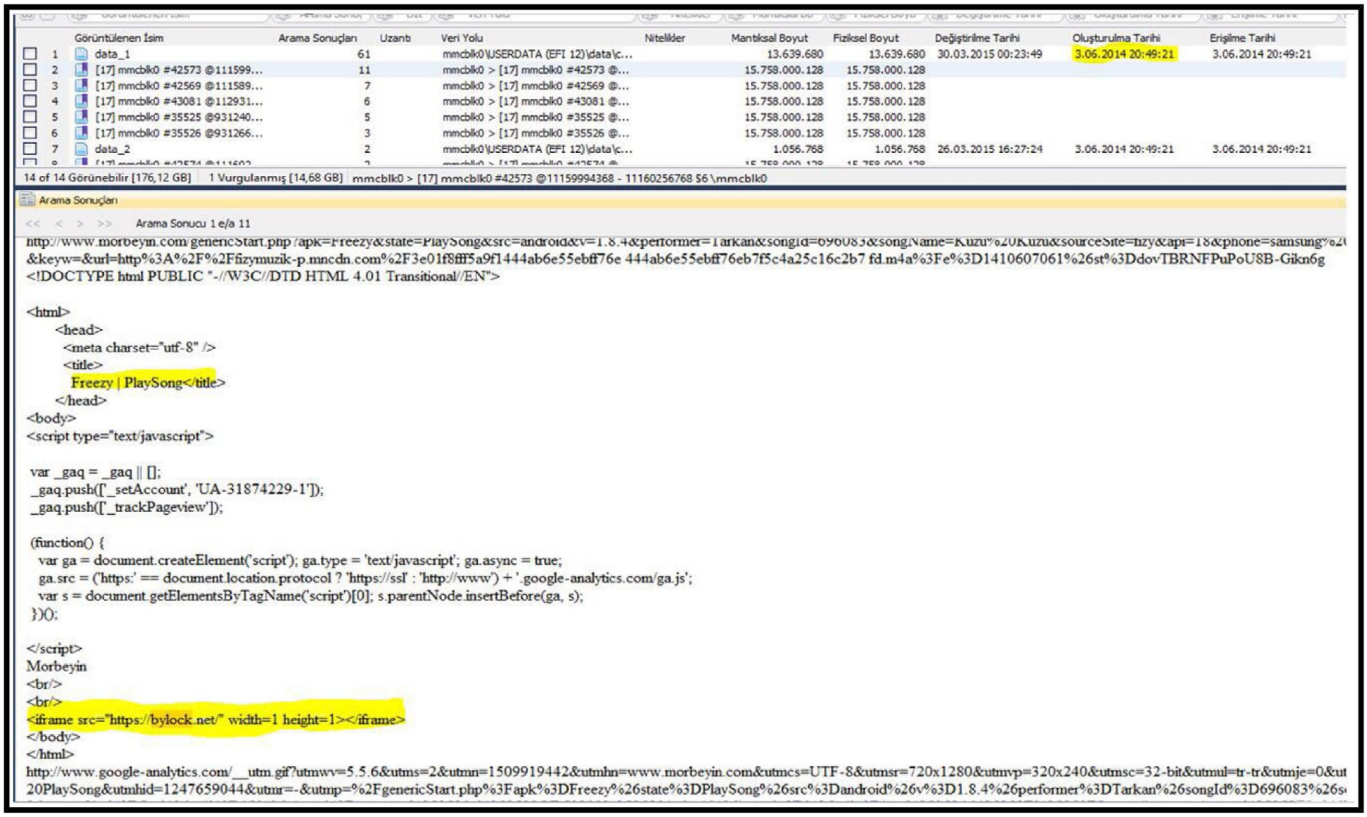


Fig. 13. A screenshot from the technical analysis report of a ByLock case defendant's mobile phone (Peksayar, 2017).

application. Moreover, the CGN and HTS records that have been provided for the purpose of this research by the defendant have flaws. If it is assumed that even as little as 1% of CGN records is inaccurate, this would result in mistrial of 1000 people on a list of 100,000 subscribers. Moreover, based on the limited data used in the research, this paper has found that the rate of flawed CGN logs is much higher.

There is another problem with the CGN shown as an example in Fig. 4. This error is not inherently due to the CGN system itself. It is instead of how the logs are evaluated and concluded in this case. Since ByLock was a chat application similar to WhatsApp (Fox-IT, 2017), it is expected to connect regularly to the remote server to check whether there are any incoming messages or not. So, it generates hundreds of sessions in short time usage. In addition to very few connections provided in the CGN file, there are not even enough connections to register the application. The connections are split into 3 day at the CGN report. There are only two connections on the first day (13 October 2014), and on the second day (8 December 2014), there is only one connection. According to Beşikçi, at least six connections are required to register and login to the application (Beşikçi, 2017b). Thus, the user of the case in this paper could not have registered the application in the first place even if all of the records were flawless. Therefore, it is a possibility that third-party apps like Freezy generate such connection logs. If users who have a low number of logs were excluded (because they could not have possibly registered or used the application), independently from CGN logs accurateness, the defendant would never get prosecuted. Nevertheless, despite these flaws and possible errors, the defendant is still on a judicial trial because his phone number was not removed from the ByLock-users list when the Morbeyin event-related list update took place.

Not enough data is publicly available to make quantitative statistical analyses of false-positive rates of reverse-scanning. Moreover, the false-positive on this matter has two different aspects. (1) The log itself can be erroneous and lead to the incorrect individual. (2) The log can be correct, but another third-party app like Freezy could have generated it. However, one can still make a logical evaluation with the data publicly available. It was revealed with the Morbeyin incident that 11,480 subscribers were deemed as false positives. Moreover, it is known that there are 95,310 people on the latest list of ByLock users (Bulur and Dirim, 2019) after the Morbeyin incident. Therefore, the false-positive rate of reverse-scan is at least $\frac{11480}{106790} = 10.75\%$. Although we cannot differentiate how much of this percent is from erroneous logs or third-party apps like Freezy. Furthermore, this ratio is still incorrect. Because the case given in Fig. 4 could not have possibly used the ByLock application due to the number of CGN sessions as explained above, and yet, it was not removed from the ByLock-users list with the Morbeyin incident. Another point worth to note is that false evidence can lead innocent people to imprisonment for years. Therefore, the validity of the evidence is exceptionally crucial.

The first list of ByLock was categorized as possible suspects with colors based on the certainty of ByLock usage (Agency, 2016b). However, we could not find any information regarding how this colors-based list was composed. Then the list of ByLock users was updated with the BTK's reverse-tracking method, and the list based on color categorization was invalidated (). With the exclusion of the users who had accession for one- or-two day, the new CGN-based list contained 122 thousand individuals. The aim of the initial exclusion of users with accession less than three days was probably to reduce the false-positive rate. Nevertheless, why the exclusion number is picked as one- or two-day and why this filtering alone is seen as sufficient is unknown, and there is no public source regarding this matter. Finally, the list was updated once again with Morbeyin.

Human Rights Council Working Group on Arbitrary Detention noted in August 2019 that using the ByLock app alone would be freedom of expression, and for prosecution, criminal activities committed by the ByLock usage have to be proven (Human Rights Council Work, 2019). Moreover, as of August 2021, there are still ByLock operations and suspects getting arrested (Kucuksahin et al., 2021) even though the European Court of Human Rights ruled on 20 July 2021 that ByLock usage alone is not evidently sufficient for pre-trial arrest (Rights, 2021).

5.2. The more reliable use of CGN logs to compose Bylock user list for prosecutions

Determining the users of the ByLock application via CGN logs could be doable if the following conditions were all met:

- The ByLock server keeps the timestamp, the public IP address, and the public port number of incoming connections.
- CGN logs provided by the ISP include the timestamp, the public IP address, and the public port number of each connection.
- The accuracy and dependability of the ISP-provided CGN logs (the model does not matter, either it is deterministic or dynamic CGN) should be evaluable and verifiable to weigh the evidential value of CGN logs correctly.

If all the above conditions were met, it could be possible to map records kept in the ByLock server to the subscribers of ISPs. Because this time, one would be auditing logs of CGN with the logs of the connected server. An excellent example of this auditing could be an individual posting illegal material on Twitter. Then authorities request the public IP address, the timestamp, and the public port of that post's session. From the data obtained, the individual can be correctly identified by using accurately kept CGN logs. However, as mentioned in the MİT's ByLock Technical Report (MİT, 2017), IP records of all incoming requests were deleted, and IP addresses belonging to Turkey were banned from accessing the server. Moreover, no public port of incoming connections was ever recorded in the ByLock server.

6. The comparison of the EncroChat and the ByLock prosecutions

EncroChat is a communication service that provides both hardware and software to its customers to provide guaranteed anonymity (E. UK, 2021). According to their official website (E. UK, 2021), the app provides the following features: Guaranteed Anonymity, Simplified Verification, Customized Android Platform, Messages That Self-Destruct, Dual Operating System, Panic Wipe, Over-The-Air Service, Password Wipe, Messaging Protocol, Secure Boot, Hardware, Tamper Proofing, Factory Restore, Updates & Live Support, Fips 140-2 Certified, and Global Service.

In July 2020, French and Dutch law enforcement and judicial authorities made a press conference to announce that they have obtained good results against cracking the EncroChat, a phone network deemed to be used mainly by criminals (Eurojust, 2020). According to the press release, encrypted phones were being widely used by organized crime groups, and French Gendarmerie and judicial authorities have been investigating them since 2017. Eventually, French authorities have put a “technical device” to bypass encryption techniques and access the users’ data. According to authorities, EncroChat was one of the biggest encrypted messaging service providers, and the majority of its user base was involved in criminal activities (Eurojust, 2020). Therefore, in 2019,

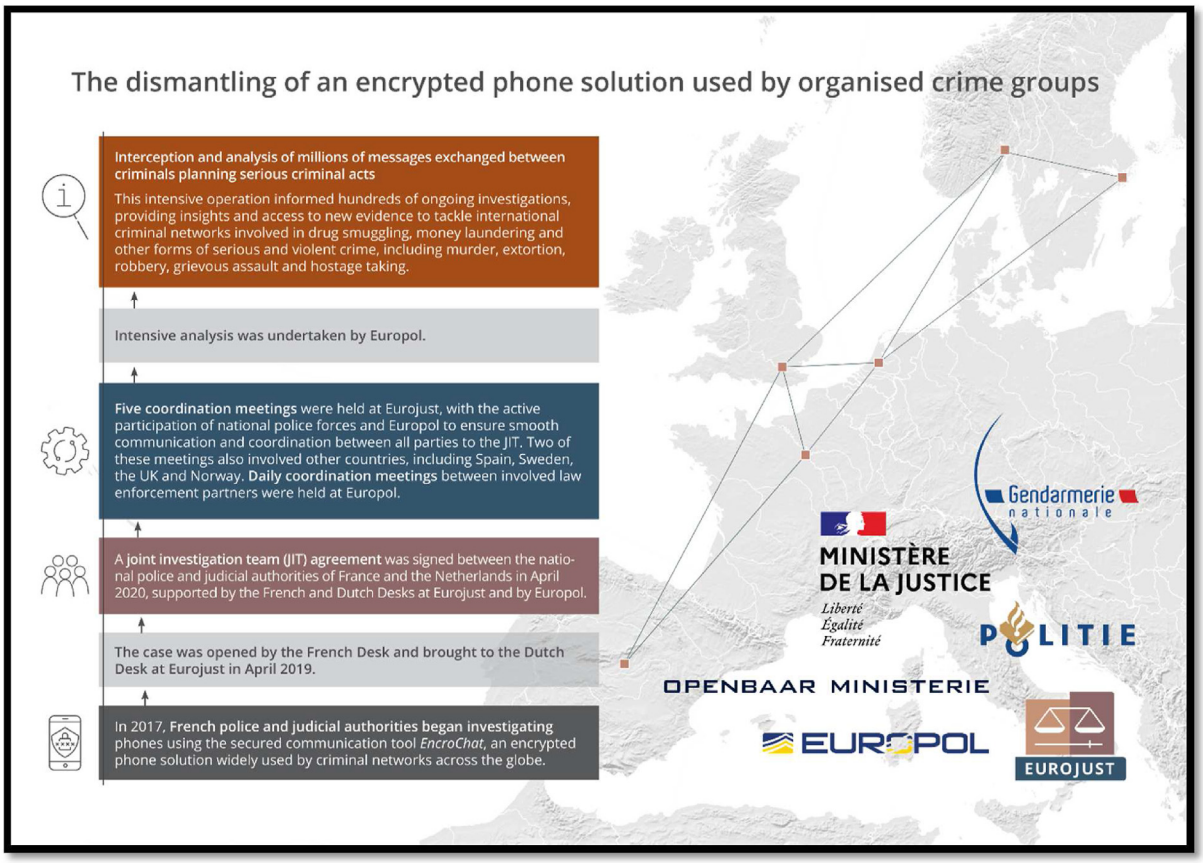


Fig. 14. The dismantling of an encrypted phone solution EncroChat used by organized crime groups (Eurojust, 2020).

French authorities opened a case at Eurojust, the EU Agency for Criminal Justice Cooperation, and joint operations against EncroChat with Europol started.

Even though authorities deemed the majority of the user base of EncroChat as criminals and EncroChat was using servers located in France (E. UK, 2021), they did not follow the technique followed by Turkish authorities in the ByLock cases. French authorities could have done a reverse scan of CGN logs to determine all users whose data subscriptions had accessed EncroChat servers like in the ByLock cases. However, they did not do such a reverse-scan to compose an EncroChat users list and arrest everyone on the list. Instead, they hacked the communication of EncroChat and captured millions of real-time messages to investigate them according to the French law and with judicial authorization through international judicial and law enforcement (E. UK, 2021). The infographic of cracking and prosecutions of EncroChat is given in Fig. 14 below. After analyses of messages, hundreds of arrests were made across Europe, and seizure of millions of euros, illegal drugs, weapons, and vehicles was made (Wright, 2020).

Although much proper methodology is followed in the EncroChat prosecutions than the ByLock prosecutions, a court in Berlin has ruled that evidence obtained from the EncroChat hack was obtained illegally (Lukas, 2021). Furthermore, the Berlin court found that this massive hacking and data collection was a severe invasion of the privacy of individuals. The court also said that EncroChat features were helpful for criminals but not much different from similar encrypted applications. This decision is vital because many similar trials are being executed in the United Kingdom, France, and the Netherlands (Lukas, 2021).

7. Conclusions

This paper has thoroughly analyzed the CGN logs usage to trackback crime suspects with actual examples from ongoing criminal cases in Turkey. It has discovered and named port-jumping errors the first time in the literature. Moreover, it has shown how reverse-tracking methodology is fundamentally flawed with the Morbeyin incident. Therefore, we believe that the evidential value of CGN logs has to be assessed very carefully, and the logs need to be used according to this value assessment in criminal cases due to the nature of CGN systems and the ISPs. Independently from local laws and regulations, due to the nature of how CGN systems work, CGN related metadata should be used carefully to start an investigation, and further action should be delicately taken. Furthermore, reverse-tracking should be used exceptionally and carefully to identify crime suspects due to the explained reasons in Section 5. Also, IPv6 deployment has to be imposed on the ISPs with stringent logging rules to obtain correct metadata. Moreover, clear and severe regulations procedures must be defined for ISPs to prevent violation of regulations.

Notice: A snapshot of all of the online references used in this research has been saved at the https://web.archive.org.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fsidi.2021.301312.

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